XIAMEN UNIVERSITY MALAYSIA

MID-TERM EXAMINATION

Course Code: CST103

Course Name: Programming language(c)

Question Paper Setter: Hamid Tahaei

Academic Session: 2020/09 Question Paper: A ☑ B ☐

Total No. of Pages: 5 Time Allocated: 2 hours

Additional Materials:

Apparatus Allowed:

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 2 Section(s):

   Section A: 2 question(s) Section B: 5 question(s)

   Answer ALL questions in section A and B
2. Read the above information carefully to ensure you have the correct and complete question paper.
3. Please follow the requirement of each section and write down all the corresponding answers on the answer book provided.
4. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly

DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO.

(Student ID:_____ Full Name:_____)

Section A (30 marks)
Correct and write Code

Question 1 (20 marks) [2 marks each]

Correct the errors in each of the following statements. (Please give the correct one on your Answer Book.)

1) Let's assume the sum of two numbers 3.1415 and 2 is 5.1415. Correct the statement below to print the correct output. Use the sum operator to calculate the sum. The output should be ("The sum of 3.1415 and 2 is 5.1415")

```
printf ("The sum of %d and %d is: \n", 3.1415+2);
```

2) Correct the following code:

```
int A12345 = 32;
if (++a12345 == 33 ) printf ("Yeahhhhhhhhh");
```

3) What is the problem with the following code? Correct it.

```
char grade;
printf( "%s", "What is your grade?\n");
scanf( "%c", grade );
printf( "Your grade is: %c\n",grade);
```

4) Correct the following piece of code:

```
int grade = 85;
Example: grade >= 60 : printf( "Passed\n" ) ? printf( "Failed\n" );
```

5) Correct the following codes:

```
int a=1, b=2;
if (a>b)
printf ("A is greater than B");
else printf ("Equal");
else if (a<b) printf ("B is greater than B");
```

6) Let's suppose that x, y and z are declared as integer. What is wrong with the following?

```
scanf ( "%d, %d, %d", &x &y &z );
```

7) What is incorrect with the following?

```
int for=0;
if (for== 5) printf ("Five");
```

8) You are required to Correct the following code using the Bitwise operator which result in 512. What should be the value of b.

```
Int a=2, b=23
printf("%d",a<<b);
```

9) Correct the following code, assuming you are going the print out every other numbers from 1 to 10. (e.g. 1, 3, 5, …)

```
for (int i= 1 ; i<=10; i++)
        printf("%d\n",i);
```

10) Correct the following:

```
Int a=1, b =2, sum;
a + b = sum;
printf(“%d”,sum);
```

Question 2 (10 marks) [2 marks each]

Write a statement for each of the following:

1) Test whether the value of a variable (e.g. count) is greater than 10. If the condition is met, print out “Count is greater than 10”.

2) Calculate the remainder of two variables and assign the result to the first variable.

3) Test whether the value of a variable (e.g. count) is greater than 5 and less than 10.

4) Using for loop and multiplication, calculate the result of a to power to b.

5) Display the n terms of harmonic series and their sum. 1 + 1/2 + 1/3 + 1/4 + 1/5 ... 1/n

Section B [70 marks]
Write Code

Question 1 [15 marks]

Write a C program to take a positive number from user (input) and determine if the number has three (3) digits or not. (e.g. print out "Ok" if the given number is 100).

Question 2 [15 marks]

Write a C program to take a positive three (3) digits number from user (input) and print out the second digit of the given number. (e.g. The second digit of the positive three digits number such as 325 is 2)

Question 3 [10 marks]

Write a C program to calculate the following equation:

$$m = \frac{a/b - c * d + e}{5}$$

Question 4 [15 marks]

Please write a C program to take an integer n from keyboard, then print out the n × n matrix. Example of the result is as following. (The program should be written using nested loop) Please follow the input and output format of the result as shown below.

```
Enter a number to print a matrix: 5
1       2       3       4       5
1       2       3       4       5
1       2       3       4       5
1       2       3       4       5
1       2       3       4       5
```

Question 5 [15 marks]

Write a C program using do-while to find HCF (Highest Common Factor) of three numbers. The greatest common factor (HCF) of a set of whole numbers is the largest positive integer that divides evenly into all numbers with zero remainder. For example, for the set of numbers 18, 30 and 42 the HCF = 6. (Factors of 18 is 2,3,6. Factors of 30, is 2,3,5,6,10,15. Factors of 42 is 2,3,6,7,14,21). Please follow the format of the result as shown below.

```
The common factor is: 1
The common factor is: 2
The common factor is: 3
The common factor is: 6
The highest common factor is: 6
```

- END OF PAPER -